

PRATYASHA BASAK

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PROFILE SUMMARY

Frontend Engineer & Data Specialist with a strong foundation in Computer Science, bridging dynamic user interfaces with data science and analytics workflows. Analytical problem-solver experienced in transforming complex analytical backends into intuitive frontend systems—from architecting modular UI data layers and geospatial visualizations to integrating predictive ML endpoints and semantic search (RAG) pipelines. Proficient in modern JavaScript/React, Python, SQL, and database management, with a track record of enforcing data integrity, optimizing frontend data ingestion, and delivering scalable applications within Agile and Scrum environments.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, SQL, JavaScript

Fullstack & Cloud (Working Knowledge): Firebase (Auth, Firestore, Hosting), REST APIs, Data Pipelines, MySQL, Data Validation & Integrity

Machine Learning & Statistics: Supervised & Unsupervised Learning, Linear & Logistic Regression, SVM, Random Forest, Decision Trees, Naive Bayes, KNN, K-Means Clustering, Regularization (Ridge & Lasso), Data Preprocessing, Pandas & NumPy, Tensorflow (Basic)

Data Visualization: Matplotlib, Seaborn, Chart.js, Apache Superset (Familiar)

Tools & Platforms: Git, GitHub, GitLab, CI/CD (basic), Firebase

Web Technologies: HTML5, CSS3, Tailwind CSS, Streamlit, React.js, Material-UI (MUI), React-Leaflet / Leaflet

Generative AI: Gemini, Claude, Google AI Studio, Chatgpt

EDUCATION

B.Tech in Computer Science and Engineering — Adamas University, Kolkata | 2023 – 2027
CGPA: **8.66**

Class XII (CBSE) — Kalyani Public School | 2023
Score: **83.4%**

Class X (CBSE) — Adamas World School | 2021
Score: **93.8%**

EXPERIENCE

Artificial Intelligence Trainee — Infosys Springboard | Oct 2025 – Dec 2025

- Implemented Python-based AI and ML workflows covering search algorithms, heuristic problem-solving, and data preprocessing pipelines, strengthening foundations in algorithmic thinking and scalable logic design.
- Applied structured data handling and modular workflow design to build end-to-end ML pipelines, reinforcing understanding of feature engineering, model evaluation, and data integrity validation.
- Collaborated on code reviews and version control workflows using Git, applying software engineering best practices to maintain code quality and system reliability across iterative development cycles.

Software Development Intern — *Harit Vikas Technologies Pvt. Ltd. | Jan 2026 – Present*

Saaya MRV Platform (Ongoing Opensource Project)

- Reorganized the platform's core React frontend to use reusable, modular components, which simplified adding new features and made the 15+ module codebase easier to maintain.
- Built a unified data adapter layer in TypeScript that connects the frontend directly to Google Firestore, REST APIs, and satellite imagery . This validated incoming data and reduced the time to integrate new data sources by ~60%.
- Created an interactive 24-hour timeline player on a React-Leaflet map. Users can play, pause, or scrub through the timeline to see smooth, step-by-step movements of tracking pointers (such as livestock) on the map.
- Developed interactive charts and map overlays that display satellite-based plant health (NDVI) data over time, helping users monitor environmental changes
- Created an automated map placement tool that automatically generates over 500 location markers on the map from coordinate datasets. This cut down manual data entry time by 90% and prevented input errors by validating data formats.
- Built a feature to upload bulk map data from Excel/CSV files and implemented group-deletion tools using Leaflet marker clusters, letting users manage thousands of map points at once while validating data for errors.
- Fixed critical package and compiler conflicts (including Webpack and TypeScript configuration targets), which improved the platform's stability and ensured updates do not break existing features

Research Intern — *IEEE SMC Student Branch Chapter, KGEC (In collaboration with IEEE SMC Kolkata Chapter & IEEE Kolkata Section) | May 2026 – August 2026*

- Selected for the highly competitive Data Science and Intelligent Analytics track (Rank 16/80; 21.05% acceptance rate from 380+ applicants nationwide); conducting research in data-driven pattern analysis and intelligent systems under IEEE SMC supervision.

PROJECTS

AI-Native Research Operating System

- Designing a hybrid semantic search and retrieval pipeline (combining BM25 keyword matching with dense vector embeddings and re-rankers) to facilitate context-aware, natural language queries across open-source scientific APIs.
- Architecting an agentic orchestration system utilizing a multi-agent planning framework—incorporating specialized Copilot, Retrieval, Synthesis, and Knowledge Graph agents—to automate complex analytical workflows.
- Developing a researcher-exclusive social media layer allowing peers to publish paper links, host AI-generated summaries, and interact through comments, likes, and professional connection maps.
- Implementing multi-author collaborative writing environments alongside private workspaces equipped with dynamic tools to organize, categorize, and draft scientific research papers.
- Integrating a structured RAG (Retrieval-Augmented Generation) pipeline paired with Neo4j graph databases to securely feed factual context into collaborative documents and verify citations.

SpotNFix — Crowdsourced Civic Issue Reporting Platform

- Built a web-based system for reporting civic issues using HTML, CSS, JavaScript, and Firebase.
- Integrated Google Maps API for auto-location and issue visualization.
- Implemented an image classification pipeline by integrating pre-trained image classification endpoints for automated content verification., duplicate detection, and issue categorization — reducing invalid report submissions and improving data integrity of the crowdsourced dataset.

DART — Decentralized Acoustic Recognition & Triangulation System

- Developed an edge-computed acoustic surveillance engine using an INT8 quantized TensorFlow Lite deep neural network to classify gunshot transients against ambient environmental noise from sliding-window Mel spectrograms.
- Implemented real-time digital signal processing (DSP) workflows in Python, applying bandpass filtering, dynamic energy thresholding, and continuous spectrogram frame buffering.
- Integrated local edge inference with a Holochain peer-to-peer Distributed Hash Table (DHT) network by constructing custom Message Pack-encoded WebSocket RPC clients.
- Built decentralized consensus and validation zones compiled to WebAssembly (WASM) to cryptographically verify multi-sensor acoustic time-difference-of-arrival (TDOA) claims against speed-of-sound constraints before DHT commitment.
- Simulated multi-node Kitsune gossip synchronization across isolated sandbox conductors to validate fault-tolerant acoustic event logging without a centralized backend.

CERTIFICATIONS

- C Programming — Great Learning Academy
- Fundamentals of Python Programming — Infosys
- The Joy of Computing with Python — NPTEL
- Artificial Intelligence: Search Methods for Problem Solving — NPTEL
- Data Analytics with Python — NPTEL

ACHIEVEMENTS

- 5 star Gold Badge in C Programming on HackerRank
- Qualifier — Smart India Hackathon (Internal) 2025 and 2026
- Clash of Coders — 2nd Round Qualifier